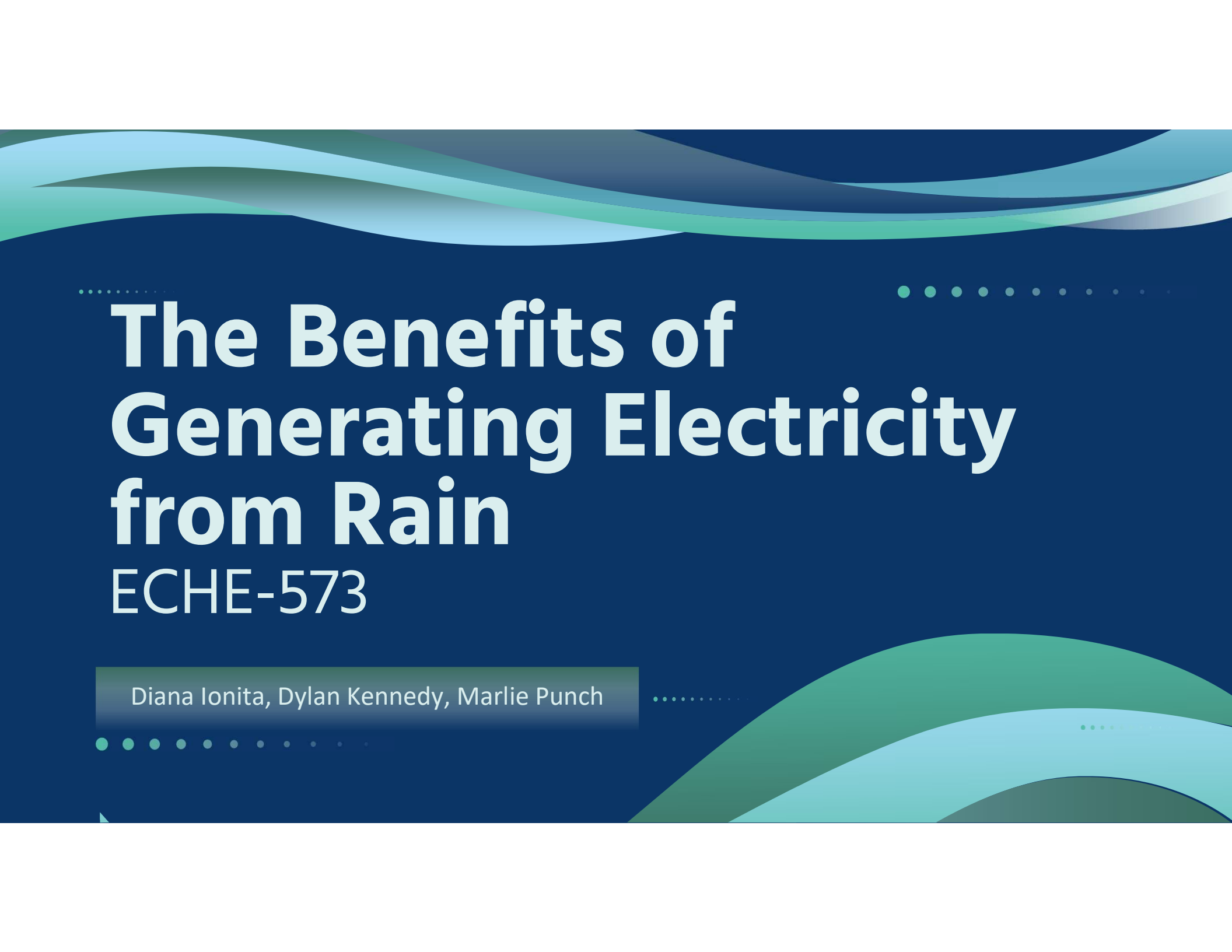




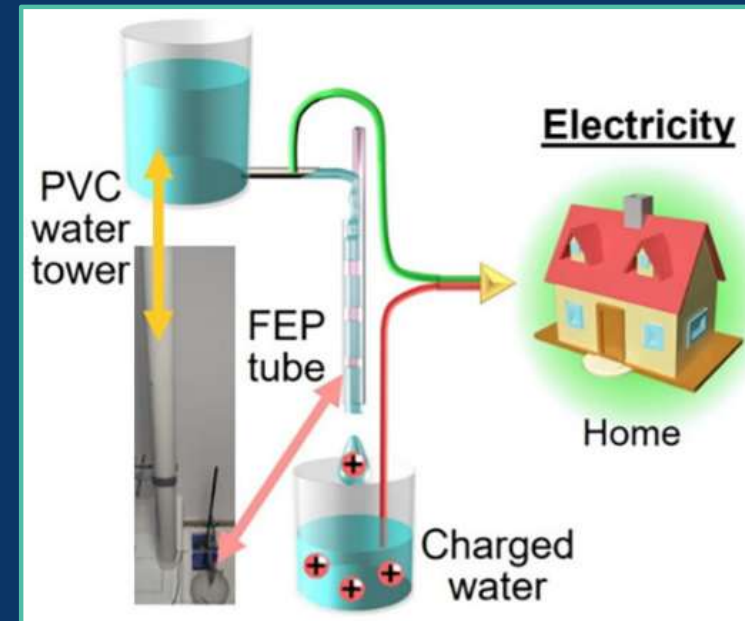
The Benefits of Generating Electricity from Rain

ECHE-573

Diana Ionita, Dylan Kennedy, Marlie Punch

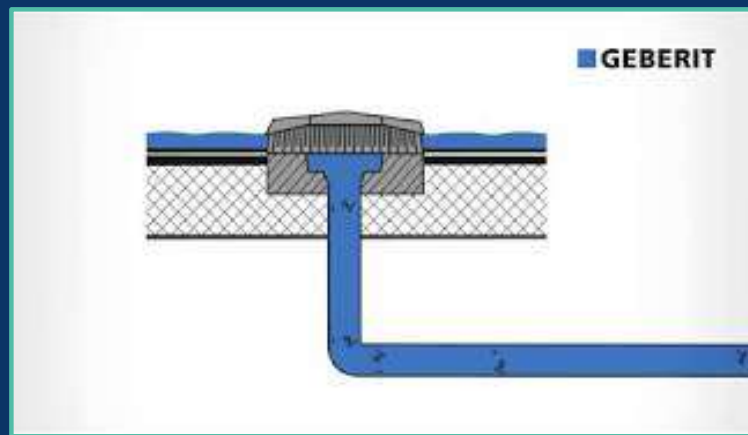
What is it?

- Uses rainwater as renewable system
- Collected & directed
- Moving water has energy



2 Main Forms of Generation:

Flow-Based

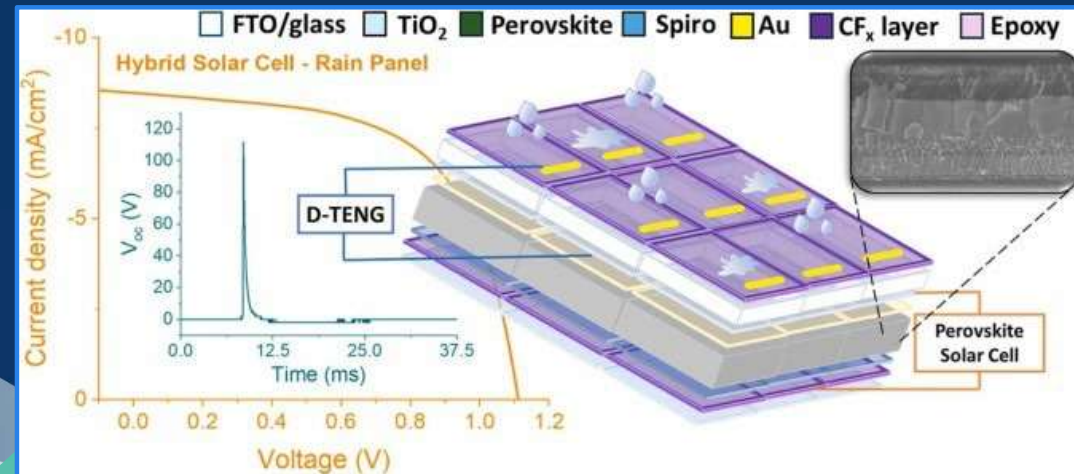


Impact-Based



Hybrid Solar Panels

- Water contact $\rightarrow \ominus$
- Surface $\rightarrow \oplus$
- Captured by Triboelectric nanogenerators (TENG)
- Charge $\rightarrow$ electricity



Why does it Matter?

- 1) Cost effective
- 2) Highly scalable
- 3) Positive environmental impacts



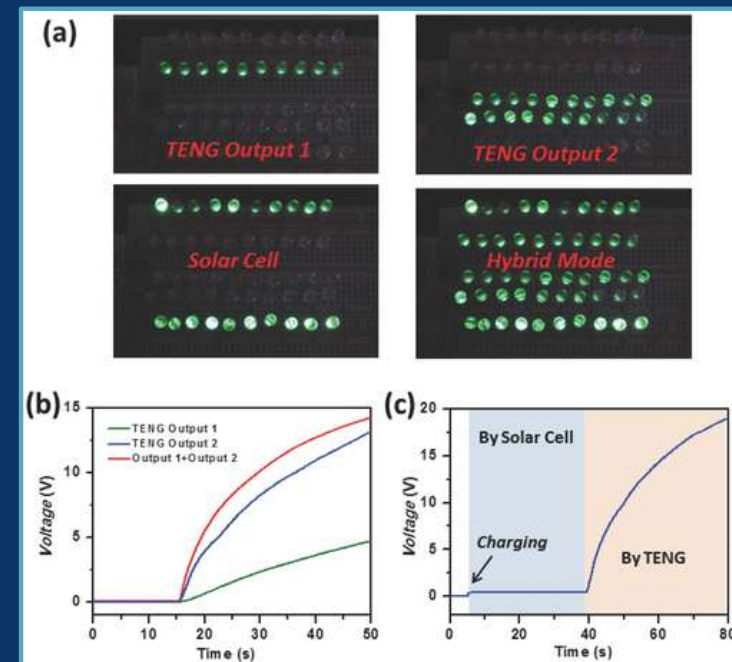


01

Cost Effectiveness

Cost Effectiveness

- Energy makes up ~25-30% of utility costs
- One system, lots of value
- \$8.7M grant in Columbia



Solar-Side Savings

Generation: $50\% \cdot 15,300 \text{ ft}^2 \cdot 14 \frac{\text{W}}{\text{ft}^2} = 107.1 \text{ kW}$

$$17.1\% \cdot 107.1 \text{ kWh} \cdot \frac{8,760 \text{ hrs}}{\text{yr}} = 160,432 \text{ kWh/year}$$

Savings: $160,432 \text{ kWh} \cdot \frac{\$0.1064}{\text{kWh/yr}} = \$17,070/\text{year}$



Rain-Side Savings

Generation: $P = (4\%)(\frac{1}{2})(1000)(3.64 \cdot 10^{-8})(9)^2(710.7) = 0.042 \text{ W}$

$$0.042 \text{ W} \cdot \frac{1 \text{ kW}}{1000 \text{ W}} \cdot \frac{8,760 \text{ hrs}}{1 \text{ yr}} = 0.366 \text{ kWh/year}$$

Savings: $0.366 \text{ kWh} \cdot \frac{\$0.1064}{\text{kWh/yr}} = \$0.04/\text{year}$



Water-Side Savings

Collection: $15,300 \text{ ft}^2 \cdot 3.77 \frac{\text{ft}}{\text{yr}} \cdot 7.481 \frac{\text{gals}}{\text{ft}^3} = 431,512 \text{ gallons/year}$

Savings: $431,512 \text{ gals} \cdot \frac{\$3.98}{1000 \text{ gals}} = \$1,717/\text{year}$

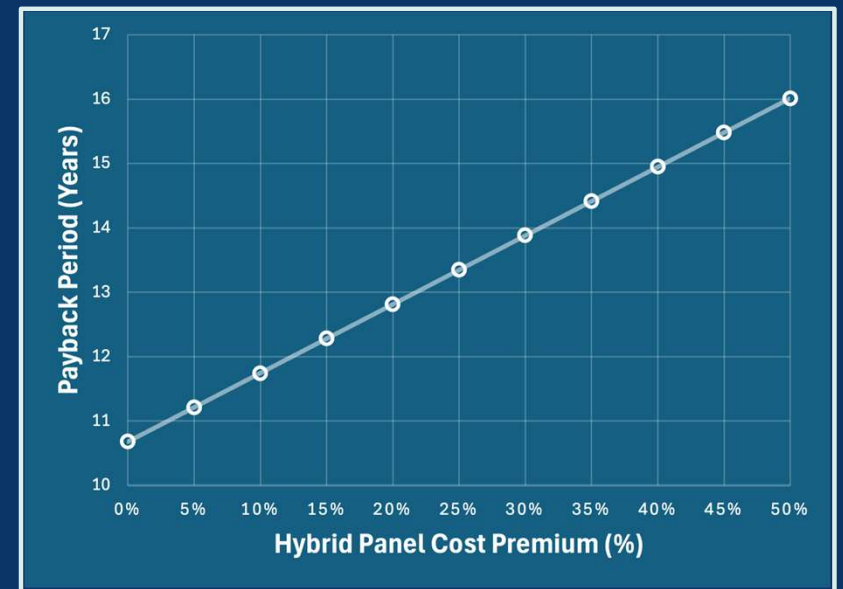


Return On Investment

Gross Savings: ~\$19,000/year

O&M: ~\$2,500/year

Net Savings: **\$16,500/year**





02

High Scalability

High Scalability

- Can be installed in phases
- Easy expansion across multiple buildings
- Useful for small- and large-scale facilities





03

Positive Environmental Impacts

Positive Environmental Impacts

- No greenhouse gasses generated
- Reduced infrastructure footprint
- Preserves local resources
- Prevents erosion



Limitations

- Supplement, not replacement
- Rainfall is variable
- Less viable for homeowners





References

A Hybridized Power Panel to Simultaneously Generate Electricity from Sunlight, Raindrops, and Wind around the Clock. (2015). *Advanced Energy Materials*.
<https://doi.org/10.1002/aenm.201501152>

City of Columbia. (n.d.). *MONDAY: EPA and Congressman Clyburn to highlight solar power conversion project at Columbia, SC wastewater treatment plant.*
<https://columbiasc.gov/monday-epa-and-congressman-clyburn-to-highlight-solar-power-conversion-project-at-columbia-sc-wastewater-treatment-plant/>

City of Columbia Water. (n.d.). *Rates.* <https://columbiascwater.net/rates/>

CleanWater2020.com. (n.d.). *Columbia goes for LEED Gold with new Water & Sewer Administration site.* <https://cleanwater2020.com/columbia-goes-for-leed-gold-with-new-water-sewer-administration-site/>

Clemson Home & Garden Information Center. (n.d.). *Best practices for application of harvested rainwater on edibles.* <https://hgic.clemson.edu/factsheet/best-practices-for-application-of-harvested-rainwater-on-edibles/>

Miller, D. C. (n.d.). *New climate normals: A comparison of the 1991–2020 climatological normals with the 1981–2010 climatological normals, focusing on Columbia, SC and Augusta, GA.* National Weather Service. https://www.weather.gov/media/cae/New_Climate_Normals_public.pdf

NASA Global Precipitation Measurement Mission. (n.d.). *How fast do raindrops fall?* <https://gpm.nasa.gov/resources/faq/how-fast-do-raindrops-fall>

National Renewable Energy Laboratory. (2024). *Commercial PV | Electricity | 2024 ATB.* https://atb.nrel.gov/electricity/2024/commercial_pv

New Atlas. (2018). *Hybrid solar cell captures energy from the sun and raindrops.* <https://newatlas.com/hybrid-solar-cell-raindrops/53713/>

U.S. Energy Information Administration. (n.d.). *Electric Power Monthly – Table 5.6.A. Average price of electricity to ultimate customers by end-use sector.*
https://www.eia.gov/electricity/monthly/epm_table_grapher.php?t=epmt_5_6_a

U.S. Environmental Protection Agency. (n.d.). *Energy efficiency for water utilities.* <https://www.epa.gov/sustainable-water-infrastructure/energy-efficiency-water-utilities>

ZME Science. (n.d.). *Hybrid solar panel turns raindrops into electricity.* <https://www.zmescience.com/science/news-science/hybrid-solar-panel-turns-raindrops-into-electricity/>